

Poland CPI Prediction Comparison with Slovenia CPI

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Abstract—This study analyzes and forecasts the Consumer Price Index (CPI) for Poland and evaluates its macroeconomic relationship with Slovenia using monthly IMF data (1990–2022). Univariate models, including ARMA on STL-decomposed residuals and SARIMA on differenced raw data, were initially developed. However, out-of-sample testing (2021–2022) revealed these backward-looking models failed to capture the unprecedented inflationary shock driven by recent global crises. To address this structural break, a multivariate Vector Autoregressive (VAR) model was implemented, leveraging the contemporaneous cross-correlation between the Polish and Slovenian economies. While the VAR model significantly outperformed the univariate approaches by anchoring the forecast to shared regional dynamics, it still underestimated the extreme magnitude of the “Black Swan” event, highlighting the limitations of historical time series modeling during unprecedented macroeconomic volatility.

Index Terms—Time Series Forecasting, CPI, ARMA, SARIMA, STL Decomposition, VAR, ADF test, ACF, PCAF, Poland, Slovenia.

I. INTRODUCTION

The Consumer Price Index (CPI) is considered one of the most important macroeconomic indicators. CPI data are of crucial importance for national governments, requiring reliable measures of price changes to make accurate economic decisions regarding wages, pensions, and inflation estimates.

In the current state of the art for macroeconomic forecasting, univariate Box-Jenkins methodologies and exponential smoothing remain foundational baselines [1]. However, to account for complex, non-linear economic dynamics and structural breaks, modern approaches heavily favor multivariate frameworks and Machine Learning techniques [1], [2]. These models leverage cross-country correlations and exogenous macroeconomic variables to capture high-inflation regimes and improve predictive robustness beyond traditional linear constraints [2].

This project aims to analyze and forecast the CPI Time Series (TS) for Poland, followed by a multivariate comparison evaluating its relationship with the CPI of Slovenia.

The dataset utilized for this study was obtained from the International Monetary Fund (IMF) and contains monthly CPI values across 191 countries. For the scope of this assignment, the target indicator is the Consumer Price Index for All Items (PCPI_IX). The data was filtered to include records from January 1990 onwards. An initial coverage verification

confirmed that both Poland and Slovenia possess 392 months of complete data (100% coverage) with no missing values.

II. METHODS

A. Time Series (TS) Splitting

To ensure robust model evaluation and prevent data leakage, the Polish and Slovenian CPI time series was chronologically split into three distinct sets:

- **Training Set:** January 1990 to December 2016 (324 months).
- **Validation Set:** January 2017 to December 2020 (48 months).
- **Test Set:** January 2021 to August 2022 (20 months).

The figure 1 below contains both TS split:

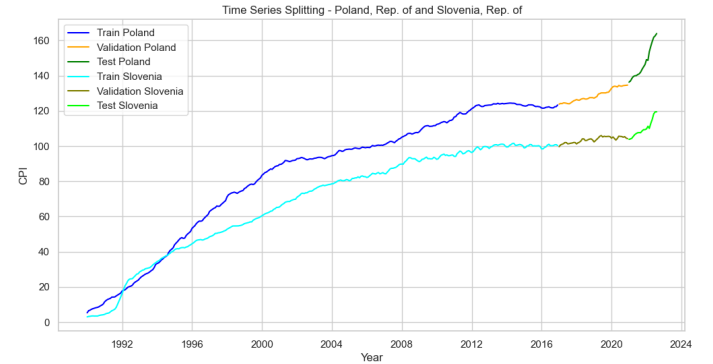


Fig. 1: Poland and Slovenia TS splits

All characterization, decomposition, and model training steps were restricted strictly to the Training Set.

B. Decomposition and Stationarity Assessment

A Seasonal and Trend decomposition using Loess (STL) was applied to the training data to isolate the underlying trend, seasonal, and residual (erratic) components as seen in figure 2.

To mathematically assess stationarity, the Augmented Dickey-Fuller (ADF) test was utilized. The STL residual component was subjected to the ADF test, yielding a highly significant statistic ($p \approx 3.15 \times 10^{-9}$), leading to the rejection

of the null hypothesis and confirming the residuals were strictly stationary. Also, we confirmed this test with the visual ACF and PACF of residuals.

TABLE I: ADF Test Result for Residuals Analysis

Metric	Value
ADF Statistic	-6.73879955051715
p -value	$3.1572462509538646 \times 10^{-9}$
Result	Stationary (Reject H_0)

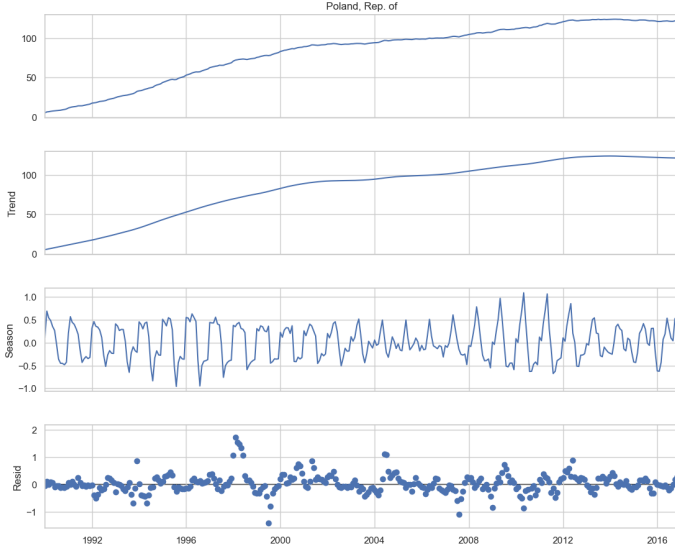


Fig. 2: Poland Original TS decomposed using STL Loess

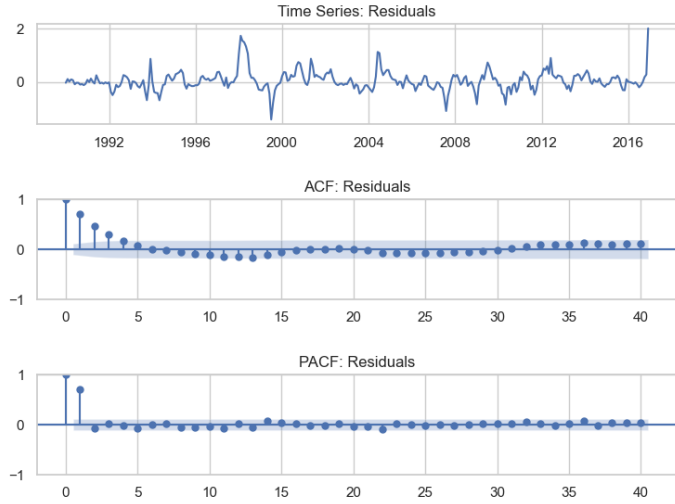


Fig. 3: Stationarity assessment through ACF/PACF

C. Univariate Model Fitting Strategies

In accordance with Box-Jenkins forecasting principles, two modeling methodologies were investigated:

a) *ARMA Modeling on Residuals*: Because the STL residuals were inherently stationary, they were modeled using a non-integrated Autoregressive Moving Average (ARMA) structure ($d = 0$). Evaluating the figure 3, we discovered that $p = 3, q = 1$, giving then an ARMA(1,3) achieved an Akaike Information Criterion (AIC) score of 12.27.

Also, A grid search evaluating parameter combinations $p, q \in [0, 6]$ determined that an ARMA(1,1) model achieved the lowest AIC score of 8.27.

We compared the empirical distribution of the ACF values against a theoretical Normal distribution $\mathcal{N}(0, 1/\sqrt{N})$ to check if the residuals of the differenced series behave like white noise. According to the figure below, we considered that both ARMA models are well defined.

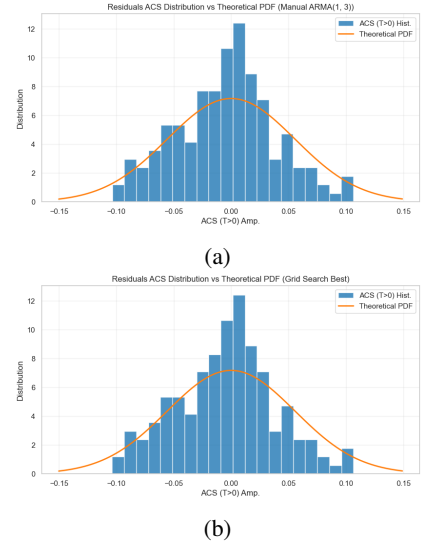


Fig. 4: ARMA Models Residual. (a)- ARMA(1,3); (b)- ARMA(1,1)

b) *SARIMA Modeling on Raw Data*: To model the raw training data directly, a Seasonal Autoregressive Integrated Moving Average (SARIMA) model was required to handle non-stationarity natively. Through iterative differencing, it was discovered that a second-order trend difference ($d = 2$) paired with a second-order seasonal difference ($D = 2$) resulted in severe over-differencing, characterized by artificial negative spikes in the Autocorrelation Function (ACF). The optimal differencing parameters were identified as first-order trend and first-order seasonal differencing ($d = 1, D = 1, S = 12$), verified in figure 5, which successfully stabilized the mean and variance, yielding a stationary ADF p -value of 1.27×10^{-5} . Visual diagnostics of the ACF and PACF at these differencing levels indicated a starting theoretical model of SARIMA(1,1,1) $\times$ (1,1,1,12).

For convenience, a grid search evaluating parameter combinations $p, q, P, Q \in [0, 3]$ determined that an SARIMA(1,1,2) $\times$ (0,1,1,12) model achieved the lowest AIC score of 169.22.

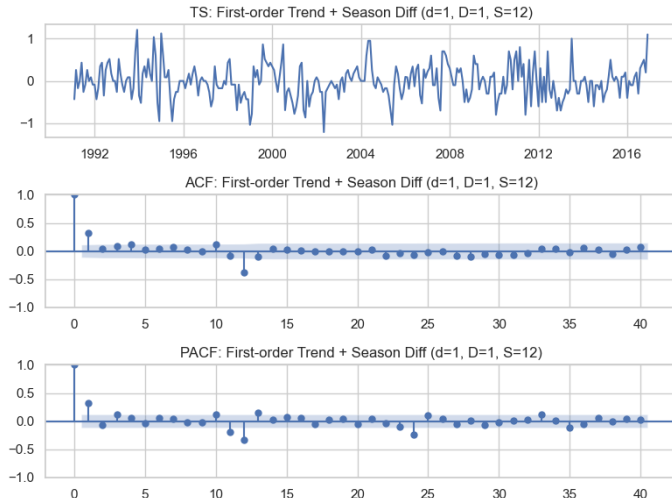


Fig. 5: Stationarity and p, q, P, Q assessment through ACF/PACF

To evaluate the definition of SARIMA models, we followed the same steps as for ARMA: We compared the empirical distribution of the ACF values against a theoretical Normal distribution. The figures above demonstrate that in fact, there are well defined.

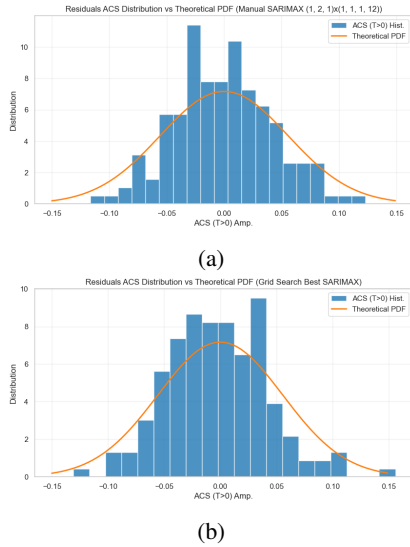


Fig. 6: SARIMA Models Residual

c) Holt-Winters (Triple) Exponential Smoothing: To perform Holt-Winters exponential smoothing, a grid search was conducted evaluating combinations of trend and seasonal components (add, mul, None). The optimal Holt-Winters model was identified with an additive trend, additive seasonality, and a seasonal period of $S = 12$.

D. Multivariate Model Fitting Strategies

In accordance with multivariate modeling requirements, the Slovenian CPI time series was evaluated. Stationarity was

achieved using the exact same differencing parameters applied to the Polish data ($d = 1, D = 1, S = 12$). This was confirmed via the ADF test and visual inspection of the ACF and PACF plots.

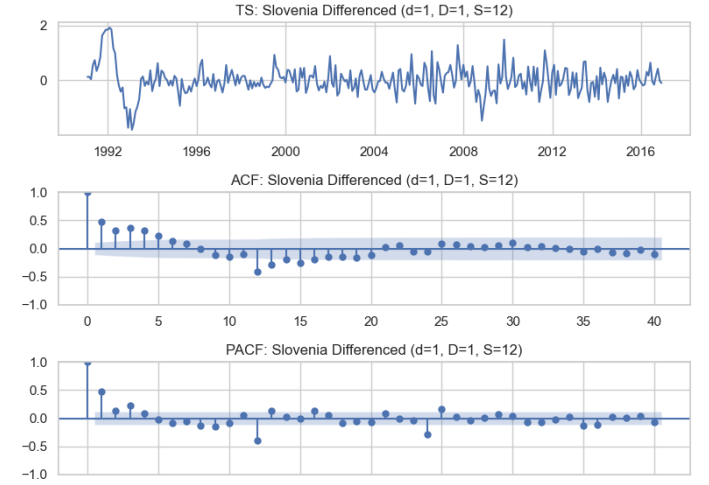


Fig. 7: Stationarity assessment through ACF/PACF

To evaluate the relationship between the two countries' time series, cross-covariance, auto-covariance, and cross-correlation sequence (CCS) analyses were performed (check the transformed TS scatter plot matrix and auto/cross-correlation matrix below). The CCS revealed a contemporaneous correlation of $< 20\%$ at Lag 0, but no significant correlations at subsequent lags. Because Slovenia's CPI does not act as a leading indicator, a SARIMAX model was deemed mathematically sub-optimal. Consequently, a Vector Autoregressive (VAR) model was selected as the simplest suitable method to capture the coupled dynamics without inducing overfitting.

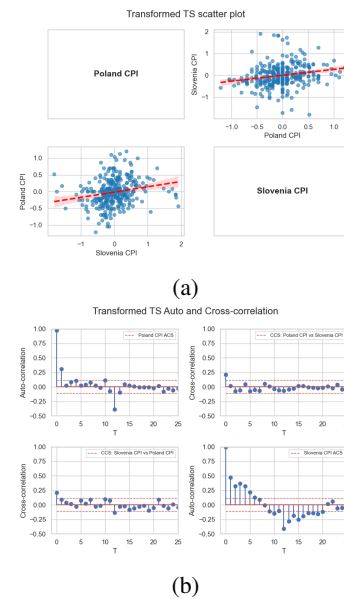


Fig. 8: Scatter plot and Auto/Cross-Correlations between TSs

III. RESULTS

All univariate and multivariate models were fitted to the training/validation data and evaluated against the test set, yielding a poor/mediocre predictive performance due to unprecedented macroeconomic shifts.

Table II and Figure 9 summarize the out-of-sample forecasting performance for all evaluated models. The ARMA models fitted to the STL residuals performed the poorest, yielding the highest RMSE (23.02) and MAPE (14.69%). Among the univariate approaches, the Holt-Winters Exponential Smoothing model performed the best (RMSE: 12.84), likely due to its heavier weighting of recent data points.

The multivariate VAR(12) model yielded the best overall quantitative results, achieving the lowest errors across all metrics (MAE: 8.16, RMSE: 11.16, MAPE: 5.26%). However, a visual analysis of the forecast charts indicates that while the VAR model significantly improved the prediction trajectory compared to the univariate models, it still struggled to fully capture the extreme exponential curve observed in the actual test set data.

TABLE II: Test Set Performance Metrics

Model	MAE	RMSE	MAPE (%)
Holt-Winters (T:add, S:add)	10.2649	12.8365	6.6874
SARIMA Manual (1,1,1)×(1,1,1,12)	12.4681	14.8167	8.1766
SARIMA Grid (1,1,2)×(0,1,1,12)	13.0980	15.3408	8.6066
ARMA Manual (1,0,3) + STL	21.9590	23.0223	14.6898
ARMA Grid (1,0,1) + STL	21.9590	23.0223	14.6898
VAR(12)	8.1660	11.1590	5.2619

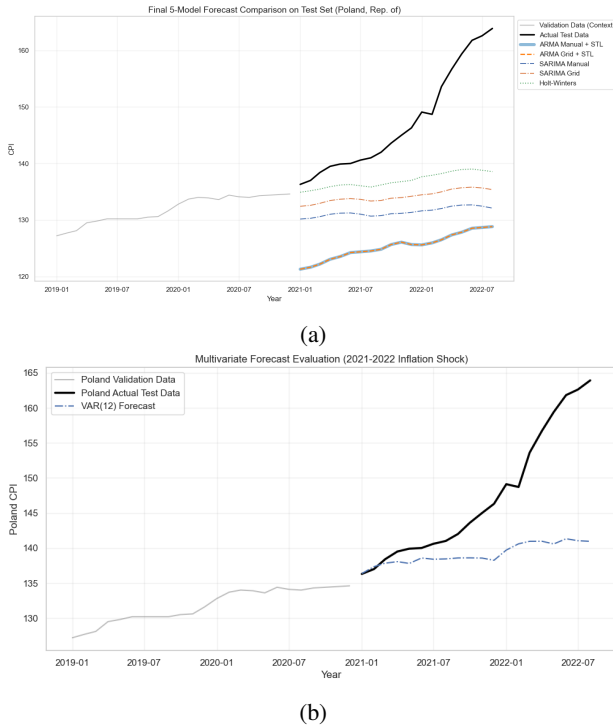


Fig. 9: Forecasting Uni/multivariate models

IV. DISCUSSION AND CONCLUSION

The forecasting results reveal a nuanced reality regarding time series modeling during structural economic breaks. The out-of-sample test set (January 2021 to August 2022) coincided with a period of severe macroeconomic volatility, driven by post-COVID supply chain bottlenecks and the geopolitical fallout of the war in Ukraine. Because standard univariate models (ARMA, SARIMA) assume future trends will strictly mirror historical data, they fundamentally failed to anticipate this spike, extrapolating a multi-decade average of low inflation into a hyper-inflationary period.

By integrating the Slovenian CPI, the VAR(12) model leveraged shared regional dynamics and successfully anchored the forecast. This approach provided a distinct predictive advantage, pulling the forecasted trajectory closer to reality and resulting in the lowest error metrics. However, as the visual analysis confirms, the multivariate approach did not perfectly mitigate the shock. The VAR model still underestimated the sheer magnitude of the unprecedented "Black Swan" event. This outcome demonstrates that while modeling the interrelatedness of coupled economies significantly helps prediction and reduces error, closed-system historical models cannot completely capture exponential, exogenous shocks without true forward-looking indicators.

A. Future Work

While this study successfully demonstrated that VAR modeling across coupled international economies improves predictive baselines, future extensions of this research should focus on high-resolution domestic forecasting. Because the CPI is fundamentally a composite basket of goods and services, the overarching inflation metric is driven by sector-specific volatility.

Future methodologies should transition from cross-country macroeconomic comparisons to domestic microeconomic predictors or a hybrid approach. By applying VAR or SARIMAX models to domestic sub-indices, such as housing costs, energy prices, and food expenditures, researchers could isolate the primary drivers of Polish inflation. This granular, sector-based multivariate approach would further improve unconditional forecasting accuracy during structural breaks and provide actionable intelligence for targeted monetary policy. Also using Rolling Time Splits for training could be more beneficial.

V. ACKNOWLEDGMENTS

This project was a key learning experience in understanding Time Series and the importance of knowing the background data domain to explain the models results. We also would like to mention that Gemini PRO (3.0 and 3.1 versions) was used to correct the report syntax, build parts of the code attached to the report and to validate our work.

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